INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

END-SEMESTER EXAMINATION 2022-23

Academic Session: 2022-2023

Programme: B.Tech. (Computer Science & Engineering) **Semester: Semester III**

Course Code: BCS-301 **Course Title: Data Structures**

Time Allowed: 3 Hours Maximum Marks: 50

Note: Attempt all sections as per instructions. Assume missing data suitably, if any.

SECTION A (Short Answer Questions - 2 Marks each)

Q1(a). Define Abstract Data Type (ADT) with an appropriate example.

Q1(b). Differentiate between Linear Search and Binary Search time complexity.

Q1(c). Explain the condition for Stack Overflow and Stack Underflow.

Q1(d). What is a Circular Queue? How does it overcome linear queue limitation?

Q1(e). Define Depth First Search (DFS) and Breadth First Search (BFS).

SECTION B (Analytical Questions - 5 Marks each)

Q2. Write an algorithm to insert and delete a node from a Doubly Linked List.

Q3. Convert the infix expression ((A + B) * C - (D - E)) ^ (F + G) into postfix form.

Q4. Construct an AVL Tree by inserting keys: 21, 26, 30, 9, 4, 14, 28, 18, 15, 10.

SECTION C (Descriptive Questions - 10 Marks each, Attempt Any Two)

Q5. Explain Quick Sort algorithm with partition logic. Analyze its Best, Average, and Worst case time complexity.

Q6. Define Minimum Cost Spanning Tree. Describe Prim’s and Kruskal’s algorithm with a graph example.

Q7. Write short notes on: (a) Collision resolution in Hashing, (b) Threaded Binary Tree.

========== END OF QUESTION PAPER ==========